

Online Appendix to “Pseudo Real-Time Nowcasting of Korean Employment: A Four-Part Test of Sentiment Signals with Sector-Scoped Evidence”

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A. OVERVIEW

Tables A1–A7 expose the numbers summarized in Sects. V–VI of the paper; section references (Sect.) below refer to the paper. They add no result. Two panel versions appear here and must not be compared across: the **pre-correction panel** precedes the industry-classification splice correction of Sect. III, and the **corrected panel** is the panel after that correction. The corrected panel exists in two builds: with the eligibility rule of Sect. III-D applied — the **adopted build**, used in Table A7; Fig. 2 of the paper additionally drops eleven inactive series — and without it, for the remaining corrected-panel results, whose recomputation on the adopted build is pending. Captions name the panel; DFM-comparison captions also name the comparison model and leg.

B. NOTES ON INDIVIDUAL TABLES

The CSI-refined failure in Table A1 is deterministic: +1.41 pp against a 1.5 pp threshold and encompassing $p=0.0753$. No placebo replication count can move either statistic.

The first row of Table A3 is the strongest claim in the paper and the easiest to overstate. It records zero verdict changes on every axis where its verdict was computed — seventeen of twenty; on the remaining three it was not computed. We write “no flips on seventeen of twenty axes, three unmeasured,” not “invariant across twenty axes.”

C. TRANSFERRING THE PROTOCOL TO OTHER SETTINGS

The pipeline is built on Korean sources, but the protocol asks for five inputs that exist in most statistical systems.

(i) **A target with a known release calendar.** A monthly official employment series, by industry if the question is sectoral, together with the date on which each reference month was first published. Snapshot masking (Sect. IV-A) needs these dates, not only the values.

(ii) **A predictor panel with an eligibility rule and a lag ledger.** Any rule serves if it is stated before results are seen; ours admits only series produced and published by public institutions or international organizations (Sect. III-D). Each series, or block of series, needs a publication lag so that a historical snapshot can hide values not yet released.

(iii) **A deterministic factor step.** Training-only standardization, PCA from a full singular value decomposition, and a median of predictions over a thin factor-count grid require no tuning and no country-specific choice. Reporting the grid region over which verdicts do not move (Table A6) replaces nominating a single factor count.

(iv) **A candidate signal and a nested pair.** The Bank of Korea sentiment indices play this role here; many central banks and statistical offices publish comparable news- or survey-based indices. The baseline and augmented models must share design and sample and differ only by the signal factor.

(v) **Enough evaluation months for the four gates.** We use at least 48 training months and an 87-month decision window. The encompassing and Clark–West gates are standard, and the placebo gate replaces only the signal channel by moving-block and phase-randomized surrogates, so it needs no external data.

Two limits carry over unchanged. Publication timing can be reconstructed from release calendars, but historical predictor values remain today’s revised values unless first-release vintages are archived for the sources in question; such archives exist for some series in some countries and not for others, so the revision layer has to be checked source by source. And a specification search that was post hoc in the original application remains post hoc: transferring the protocol does not make an explored specification prespecified.

TABLE A1

DESIGNATED FOUR CELLS, RECONSTRUCTED LATEST-AVAILABLE SCOPE. PRE-CORRECTION PANEL (*BEFORE* THE SECT. III SPLICE CORRECTION), DECISION WINDOW ($n=87$), MEDIAN $\kappa\{5,6,7\}$. RMSE IN THOUSANDS OF PERSONS; NET INCREMENT IN PERCENTAGE POINTS. THESE STORED VALUES ARE **SUPERSEDED** PENDING THE FROZEN-PANEL WHOLESALE RECOMPUTATION AND ARE NOT RELABELED AS CORRECTED-PANEL RESULTS. ON THE CORRECTED PANEL THE FIRST ROW MOVES TO PUBLIC 18.308, PUBLIC+SIGNAL 17.529, NET INCREMENT +3.956 PP. CORRECTED-PANEL GATE COUNTS BY GRID ARE IN TABLE A6. THE AR ANCHORS ARE ARM-INVARIANT.

Cell	AR(1)	AR(1,12)	public	public+signal	net incr.	enc. p	CW p	verdict
NSI $\times$ IT $\times$ refined	19.14	18.76	18.24	17.77	+2.38	0.0013	0.0021	4 gates
NSI $\times$ IT $\times$ full	19.14	18.76	18.68	18.28	+2.03	0.0045	0.0033	4 gates
CSI $\times$ IT $\times$ refined	19.14	18.76	18.25	17.98	+1.41	0.0753	0.0432	fails ①, ②
CSI $\times$ IT $\times$ full	19.14	18.76	18.67	18.21	+2.36	0.0187	0.0252	4 gates

TABLE A2

FORMAL-DFM COMPARISON ON BOTH LEGS. THE ADOPTED COMPARISON MODEL FAVORS US ON BOTH LEGS, WHILE THE MODEL IT REPLACED ATTAINS THE LOWEST RMSE IN EVERY MEASURED COMBINATION. ADOPTED COMPARISON MODEL IS THE KALMAN TWO-STEP DFM WITH A VECTOR-AUTOREGRESSIVE VAR(1) STATE (DOZ-GIANNONE-REICHLIN); THE PREVIOUS MODEL IS THE PCA-PLUS-AUTOREGRESSION TWO-STEP IT REPLACED. CORRECTED PANEL, DECISION WINDOW ($n=87$), MEDIAN $\kappa\{5,6,7\}$. POSITIVE MEANS OUR MODEL IS THE MORE ACCURATE. THE PUBLIC LEG GIVES THE COMPARISON MODEL PUBLIC FACTORS ONLY; THE SYMMETRIC LEG ALSO GIVES IT THE SIGNAL. THE REPLACED MODEL ATTAINS THE LOWEST RMSE IN 64/64 MEASURED COMBINATIONS OF CELL, LEG, GRID AND WINDOW.

Cell	signal RMSE	Public leg: vs			sym. DFM RMSE	Symmetric leg: vs	
		adopted	previous	adopted		previous	
NSI $\times$ IT $\times$ refined	17.529	+4.85%	+0.98%	17.739	+1.18%	−3.22%	
NSI $\times$ IT $\times$ full	18.019	+2.34%	−0.59%	18.200	+0.99%	−3.41%	
CSI $\times$ IT $\times$ refined	17.900	+2.87%	−1.13%	18.015	+0.64%	−2.67%	
CSI $\times$ IT $\times$ full	18.171	+1.53%	−1.42%	18.229	+0.32%	−2.33%	
in our favor		4 of 4	1 of 4		4 of 4	0 of 4	

TABLE A3

VERDICT ROBUSTNESS BY CELL ACROSS TWENTY PERTURBATION AXES. CORRECTED PANEL EXCEPT SOME PRE-CORRECTION AXES. “HELD” COUNTS AXES ON WHICH THE CELL’S THREE-GATE VERDICT DOES NOT CHANGE; DENOMINATORS DIFFER BY CELL. UNMEASURED AXES ARE SILENT, NOT FAVORABLE. “FLIPS” COUNTS INDIVIDUAL VERDICT CHANGES, SO ONE AXIS WITH SEVERAL SETTINGS CAN CONTRIBUTE MORE THAN ONE. CELLS DESIGNATED PRIMARY IN THE UNREGISTERED INTERNAL ANALYSIS DRAFT ARE MARKED $\checkmark$.

Cell	Int. primary	Axes held	Flips
NSI $\times$ IT $\times$ refined	$\checkmark$	17 / 20	0
NSI $\times$ IT $\times$ full		12 / 19	5
NSI $\times$ M $\times$ full		11 / 18	3
CSI $\times$ M $\times$ full		10 / 18	8
NSI $\times$ M $\times$ refined		10 / 19	8
CSI $\times$ IT $\times$ refined		8 / 20	36
CSI $\times$ M $\times$ refined	$\checkmark$	7 / 19	16
CSI $\times$ IT $\times$ full	$\checkmark$	7 / 19	17

TABLE A4

FOUR FACTOR-COUNT SPECIFICATIONS, SAME DATA, SAME GATES. ALL COLUMNS ARE ON THE PRE-CORRECTION PANEL; TABLE A6 REPORTS TWO OF THESE GRIDS ON THE CORRECTED PANEL AND DIFFERS. AMONG THE FOUR CANDIDATES, RULE-STABILITY RUNS OPPOSITE TO THE IT, vs-DFM AND BH COLUMNS; THE M COLUMN IS NOT. **NONE OF THESE IS “THE CURRENT SPECIFICATION”** — WE DO NOT NOMINATE ONE. **THE vs-DFM COLUMN IS SUPERSEDED:** IT IS COMPUTED AGAINST THE PREVIOUS COMPARISON MODEL, ON THE PUBLIC LEG, ON THE PRE-CORRECTION PANEL, AND WAS NOT RECOMPUTED UNDER THE ADOPTED MODEL. IT IS RETAINED BECAUSE DELETING AN UNFAVORABLE COLUMN WOULD BE WORSE. THE BH COLUMN IS COMPUTED WITHIN THE FOUR DESIGNATED IT CELLS OF EACH ROW, A DIFFERENT FAMILY FROM THE 68-CELL SECTOR SCAN OF TABLE III OF THE PAPER.

Spec.	Grid	Rule-statability	IT	M	Apparent sector	vs DFM (superseded)	BH $q \leq .05$
spec1	{1,...,6}	highest — “everything up to the scree boundary”, no further choice	0/4	1/4	M only	4/4 negative	0/4
spec2	{2,...,6}	high	0/4	2/4	M only	4/4 negative	0/4
spec3	{3,4,5,6}	high	0/4	0/4	none — qualification vanishes	2/4 negative	2/4
spec4	{5,6,7}	lowest — not statable; sweep artifact	3/4	0/4	IT only	4/4 positive (one tie)	3/4
(ref456)	{4,5,6}	partial	4/4	1/4	IT + M	positive (two ties)	4/4

TABLE A5

ACCURACY LAYER: HOW LOW THE ERROR CAN GO, AND AT WHAT SELECTION COST. RECONSTRUCTED LATEST-AVAILABLE SCOPE, THE PRE-CORRECTION PANEL (BEFORE THE SECT. III SPLICE CORRECTION). THIS LAYER DELIBERATELY RETAINS THE *PREVIOUS* EXTRACTOR: IT ASKS THAT QUESTION RATHER THAN WHICH COMPARISON MODEL THE METHOD SECTION DESCRIBES, AND THE TWO ARE NOT RECONCILED HERE. THE ROW MARKED $\dagger$ IS A MIXED SPECIFICATION (TWO MEMBERS AT MEDIAN $\{5,6,7\}$; ONE TIME-SERIES FOUNDATION-MODEL MEMBER CARRIES NO k) AND MUST NOT BE COMPARED ACROSS k . ONE FURTHER ROW WAS RETRACTED AFTER A COLUMN-ORDER SENSITIVITY AUDIT AND IS OMITTED; ITS VALUE IS NOT REPORTED. **NONE OF THESE ROWS IS USED FOR ANY GATE OR p -VALUE CLAIM.**

Layer	Configuration	k spec.	RMSE	Selection space
fixed	linear public factors + signal	median $\{5,6,7\}$	17.773	none (stored specification)
fixed	two-step DFM + signal	median $\{5,6,7\}$	17.548	none (k -spec. transplanted)
prior	two-step DFM + signal, raw	single $k=7$	17.660	—
searched	two-step DFM + signal at lag 1	single $k=7$	17.148	8 transforms $\times$ 2 k -specs $\times$ 8 cells = 128
searched	two-step DFM + signal at ma3	median $\{5,6,7\}$	17.433	same
exhaustive	two-step DFM + signal at lag 1	mean $\{5,6,7\}$	17.136	$11 \times 5 \times 5 \times 8 = 2,200$
fixed	simple mean of five extractors	median $\{5,6,7\}$	17.540	none
fixed	correlation-filtered combination of 55 members, rule written before running	median $\{5,6,7\}$	17.587	none (rule pre-declared)
post-hoc	best member/aggregation combination	mixed $\dagger$	16.916	10,250 rows

TABLE A6

THE VERDICT PLATEAU AND ITS BOUNDARY. THE FOUR-GATE COUNT DOES NOT MOVE INSIDE THE MEASURED REGION, AND THE REGION ITSELF REMAINS POST HOC. SAME DATA, SAME GATES, CORRECTED PANEL, PLACEBO $n=1,000$. “IDENTICAL” COMPARES THE EIGHT-CELL VERDICT VECTOR AGAINST $\{5,6,7,8\}$ CELL BY CELL. THE DECISION-WINDOW PLATEAU CONDITION IS: EXCLUDE $k=4$ AND REACH $k \geq 8$. ITS UPPER BOUNDARY BEYOND $k=9$ IS UNMEASURED. REFERENCE-WINDOW COUNTS ARE LOWER BOUNDS: UNRUN PLACEBOS COUNT AS FAILURES.

Grid	Mean k	4-gate, dec.	4-gate, ref.	Identical?
$\{4,5,6\}$	5.00	1/8	1/8	no
$\{5,6,7\}$	6.00	4/8	5/8	no
$\{4,5,6,7,8\}$	6.00	4/8	5/8	no
$\{6,7,8\}$	7.00	5/8	5/8	(IT 4/4)
$\{5,6,7,8\}$	6.50	5/8	5/8	(baseline grid)
$\{5,6,7,8,9\}$	7.00	5/8	5/8	yes, both

TABLE A7

RELEASED-EMPLOYMENT SURPRISE RESIDUALIZATION. ALL EIGHT COMPARISONS RETAIN ALL FOUR GATES AFTER THE RELEASED-EMPLOYMENT SURPRISE IS REMOVED FROM THE SIGNAL. THIS IS THE NON-EQUIVALENT ABLATION OF SECT. V-C. SCOPE: THE RELEASE-MASKED +33 SNAPSHOT ON THE ADOPTED BUILD OF THE CORRECTED PANEL, NOT THE RECONSTRUCTED LATEST-AVAILABLE PANEL. AT EACH EXPANDING ORIGIN THE PRIOR MONTH’S AR(1,12) SURPRISE IS FORMED FROM EARLIER MONTHS ONLY AND NSI IS RESIDUALIZED AGAINST IT WITHIN THE FOLD; THE FORECAST TARGET IS NEVER USED BY THE RESIDUALIZER. THIS REJECTS LINEAR REPACKAGING OF THE RELEASED SURPRISE. IT CANNOT SHOW THAT EAPS REPORTING NEVER ENTERS NSI AT ARTICLE LEVEL, NOR IDENTIFY A CAUSAL MECHANISM.

Cell	Grid	Window	Public	Purged +NSI	Incr.	Enc. p	CW p
NSI $\times$ IT $\times$ refined	$\{5,6,7\}$	decision	17.590	16.579	+5.135	0.000301	0.000829
NSI $\times$ IT $\times$ refined	$\{5,6,7\}$	reference	18.880	17.507	+6.344	0.000002	0.000005
NSI $\times$ IT $\times$ refined	$\{5,6,7,8\}$	decision	17.432	16.491	+4.782	0.000348	0.001082
NSI $\times$ IT $\times$ refined	$\{5,6,7,8\}$	reference	18.689	17.370	+6.095	<0.000001	0.000002
NSI $\times$ IT $\times$ full	$\{5,6,7\}$	decision	17.457	16.670	+3.998	0.001419	0.001665
NSI $\times$ IT $\times$ full	$\{5,6,7\}$	reference	18.825	17.525	+6.008	0.000002	0.000007
NSI $\times$ IT $\times$ full	$\{5,6,7,8\}$	decision	17.446	16.621	+4.193	0.000903	0.001575
NSI $\times$ IT $\times$ full	$\{5,6,7,8\}$	reference	18.775	17.486	+5.958	0.000002	0.000007